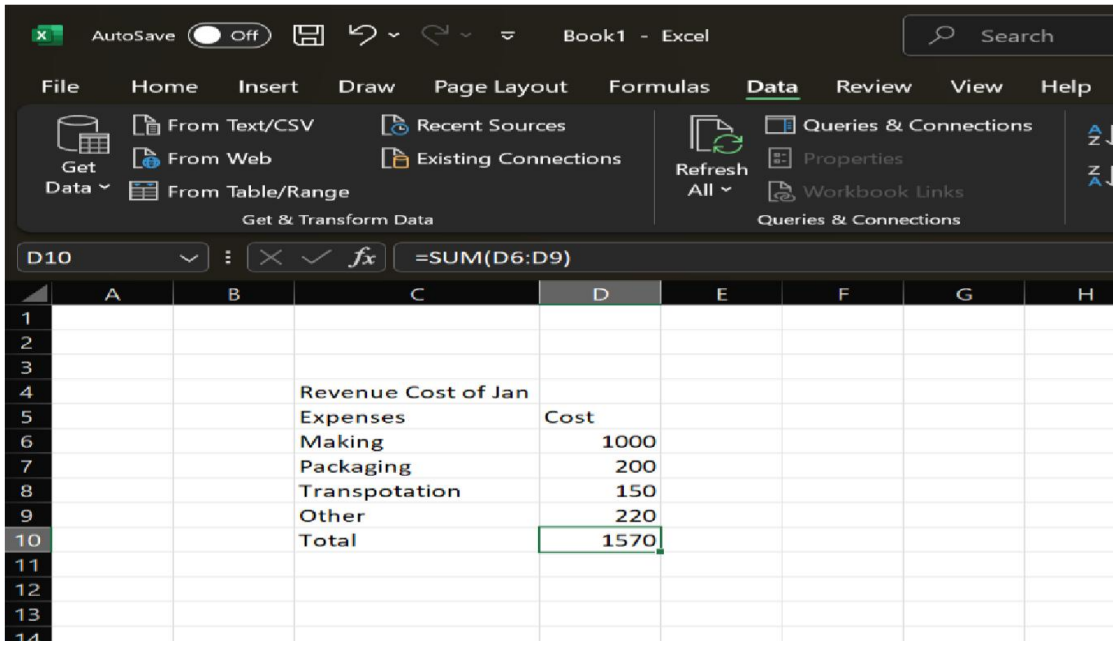
Practical 2

Apply the what – if Analysis for data visualization. Design and generate necessary reports based on the data warehouse data. Use Excel.

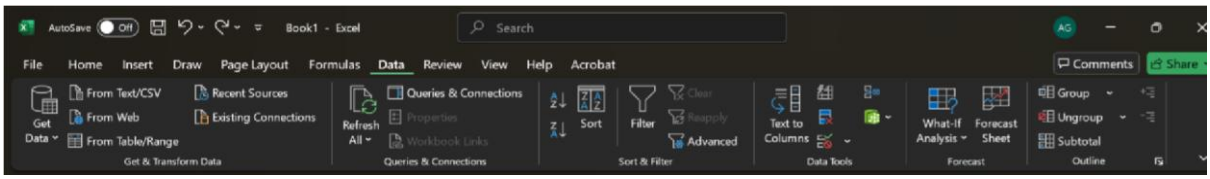
❖ Scenario Manager

In **scenario manager**, we create different **scenarios** by proving different input values for the same **variable** than by comparing scenarios to choose the correct result. For Example, To check the **cost of revenue** for **three different months.**

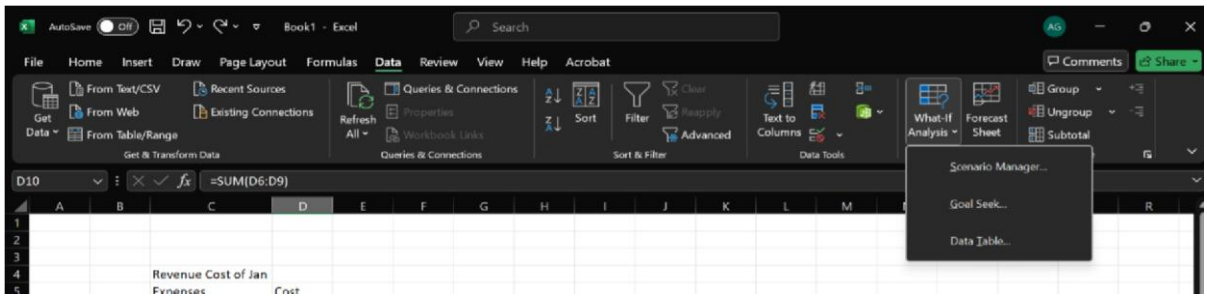
Step 1: Given a data set, for **Revenue Cost of Jan**, with **Expenses** and **Cost** as its Columns.



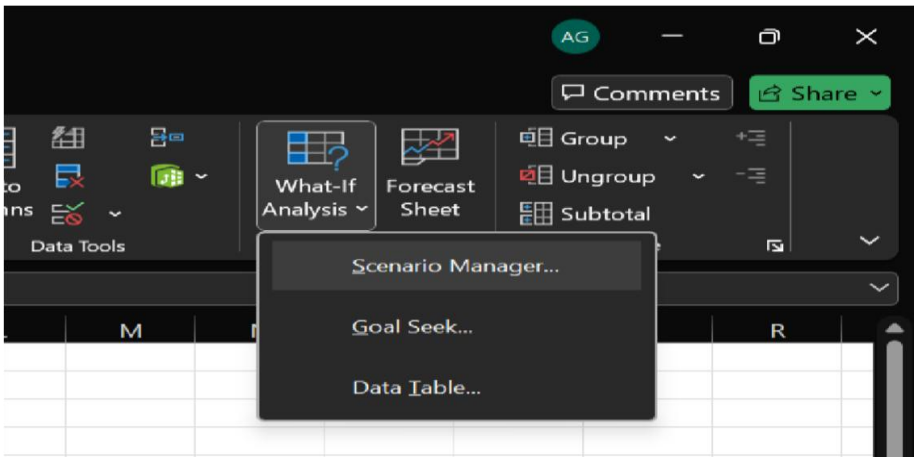
Step 2: Select the numerical value cell and Go to the **Data.**



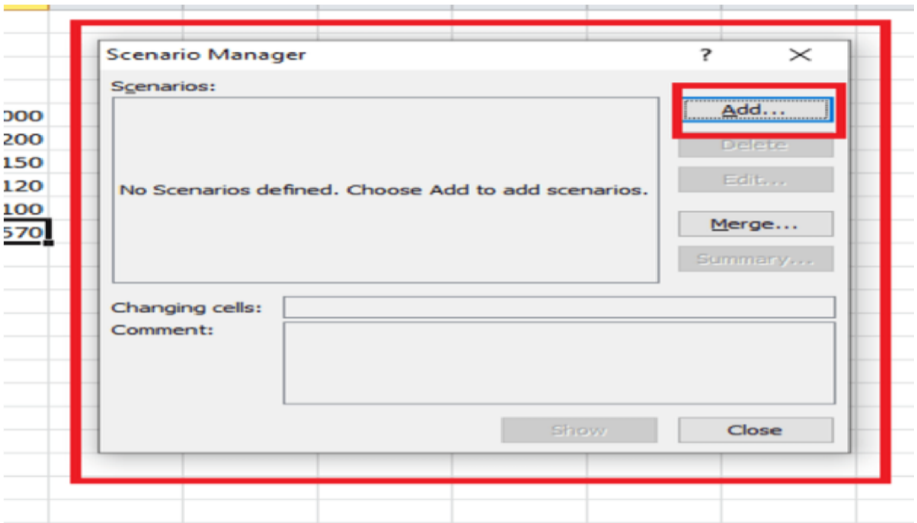
Step 3: Under the forecast section, click on the **What-if analysis**.



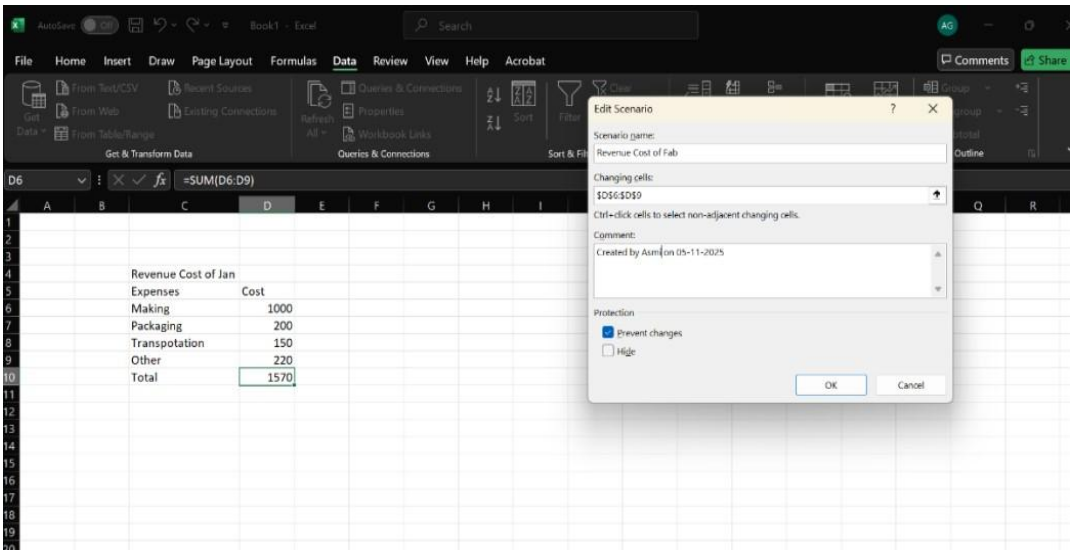
Step 4: A drop-down appears. Select the **Scenario manager.**



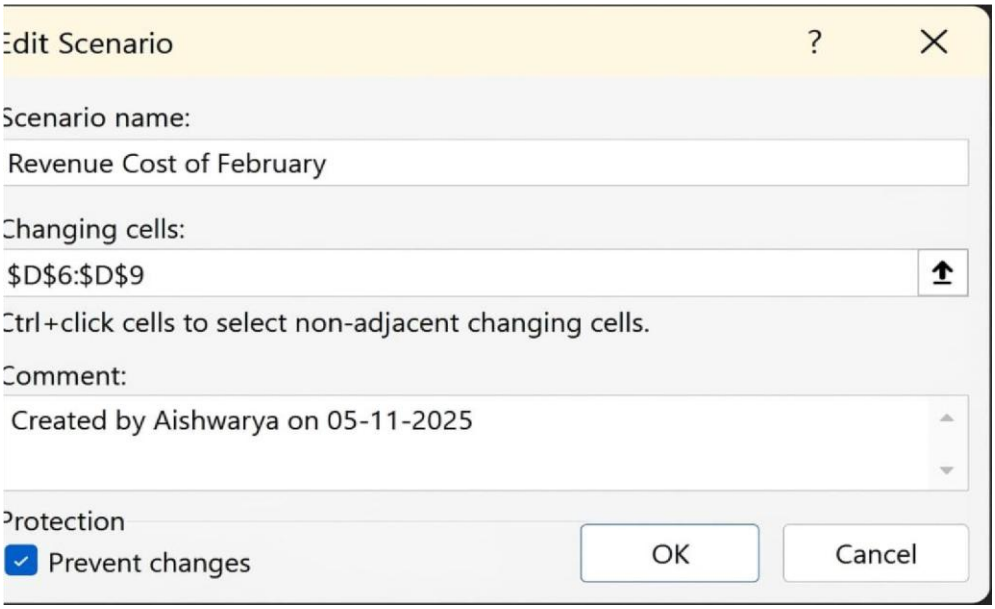
Step 5: A dialog box appears in the dialog box select **add option.**



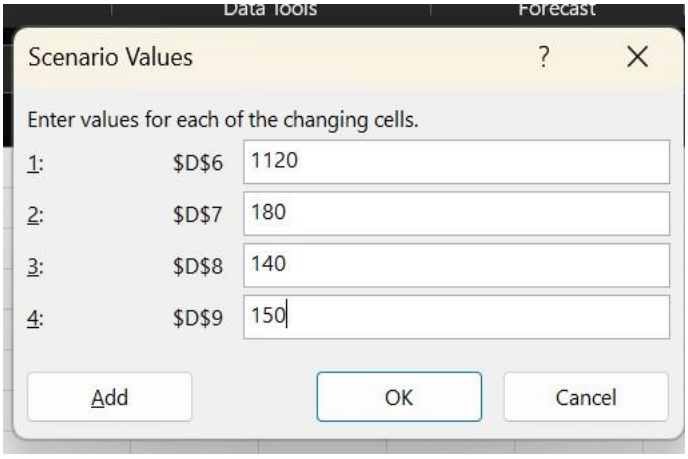
Step 6: A new dialog appears to write the name of the **new scenario** in the first column. Under **Scenario name**, write **"Revenue of Feb"**.

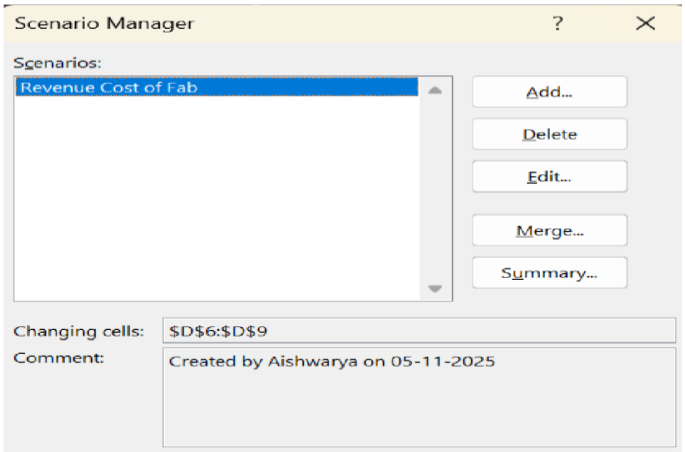


Step 7: In the second column **select the changing cell.** The **changing cells** for this example, are **$D$6:$D$9**.

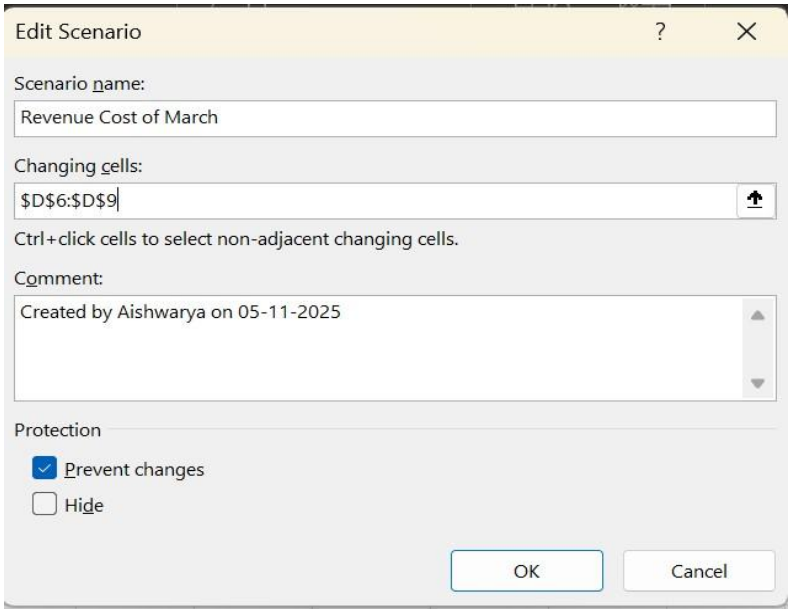


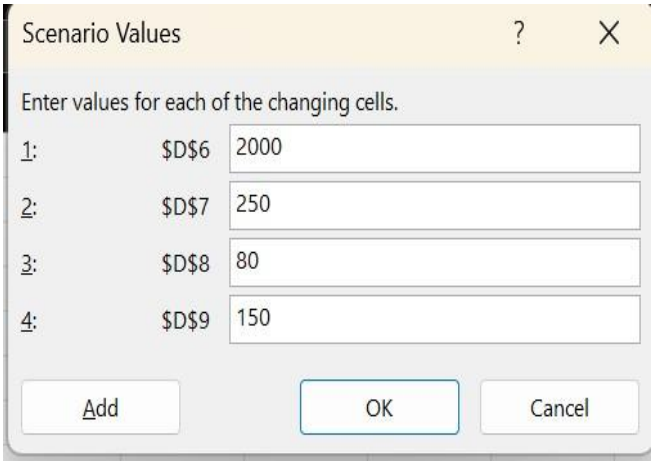
Step 8: A new dialogue box name **Scenario Values** appears to write the changed value in the box. Enter the values as per shown in the image. Click OK



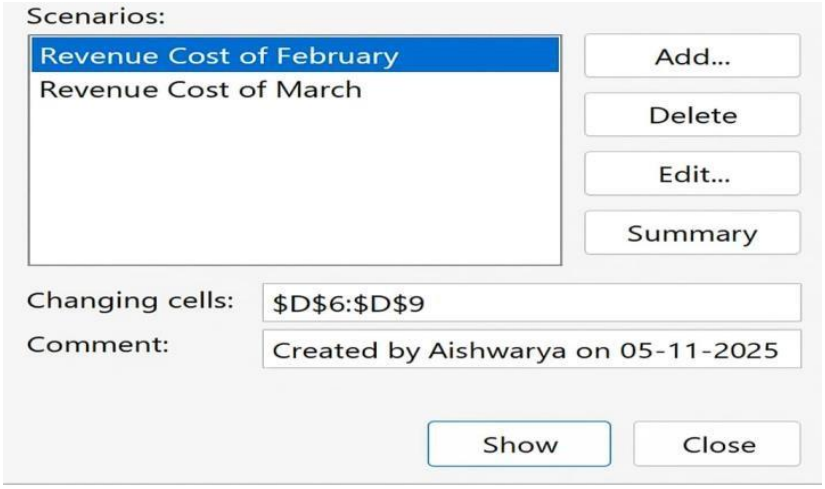


Step 9: Repeat Step 5, Step 6, Step 7 & Step 8.

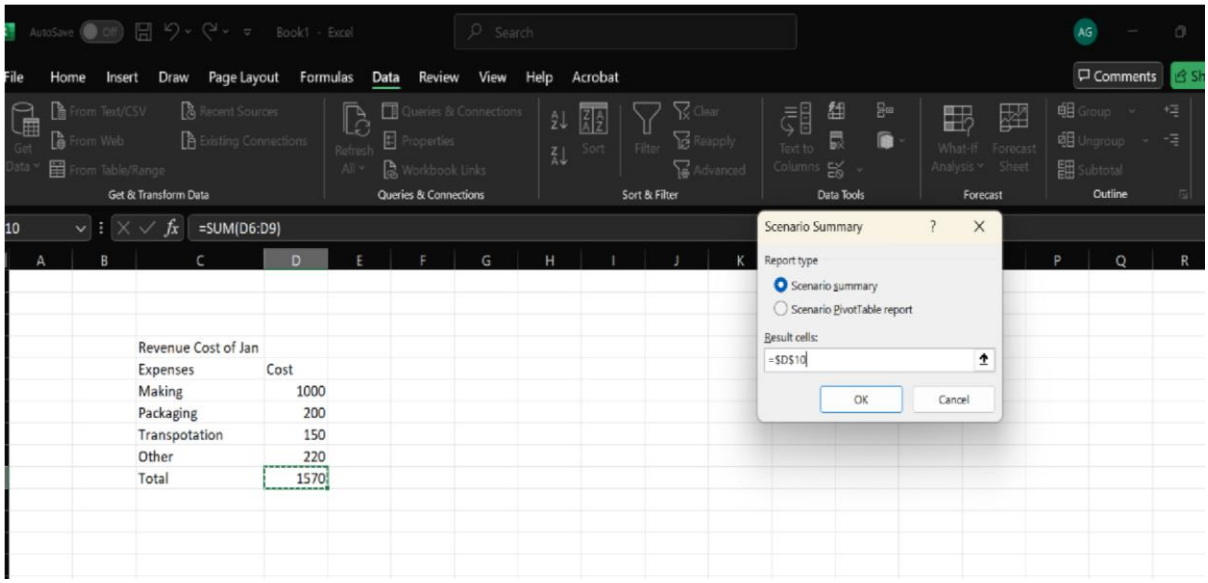




Step 10: Click **Ok** then select **summary.**



Step 11: A new Dialog box name **Scenario Summary** appears. Select **Result cells**: **$D$10**.



Step 12: See the result.

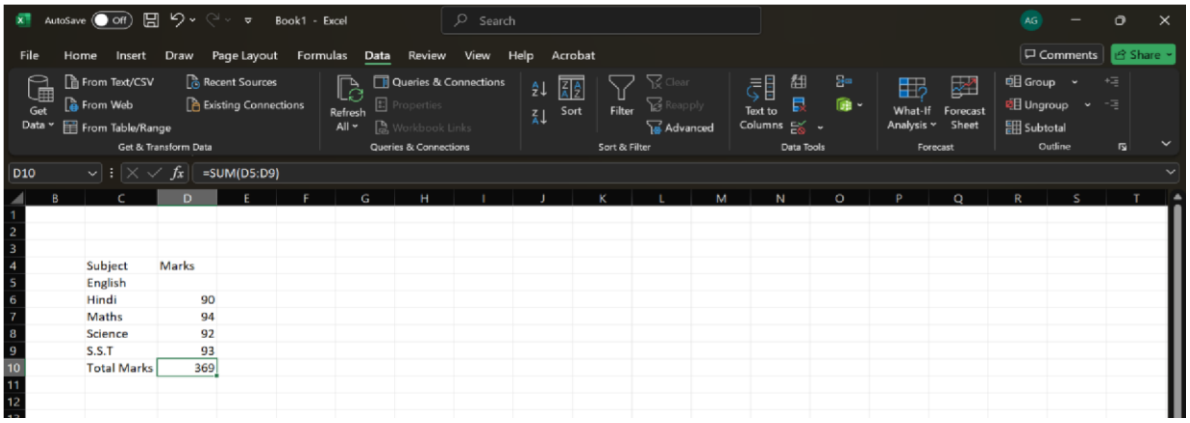
Scenario Summary	Current Values:	Revenue Cost of Fab	Revenue Cost of March
Changing Cells:			
$D$6	1000	1120	2000
$D$7	200	180	250
$D$8	150	140	80
$D$9	220	150	150
Result Cells:			
$D$10	1570	1590	2480

Notes: Current Values column represents values of changing cells at time Scenario Summary Report was created. Changing cells for each scenario are highlighted in gray.

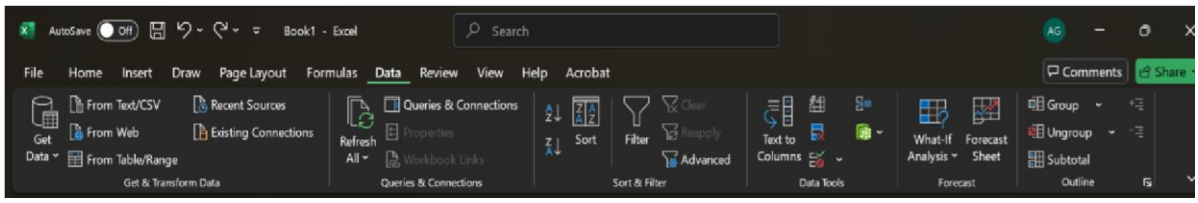
❖ Goal seek

In goal seek we already know our output value we have to find the correct input value. For example, if a **student** wants to know his **English marks** and he knows all the rest of the marks and total marks in all subjects.

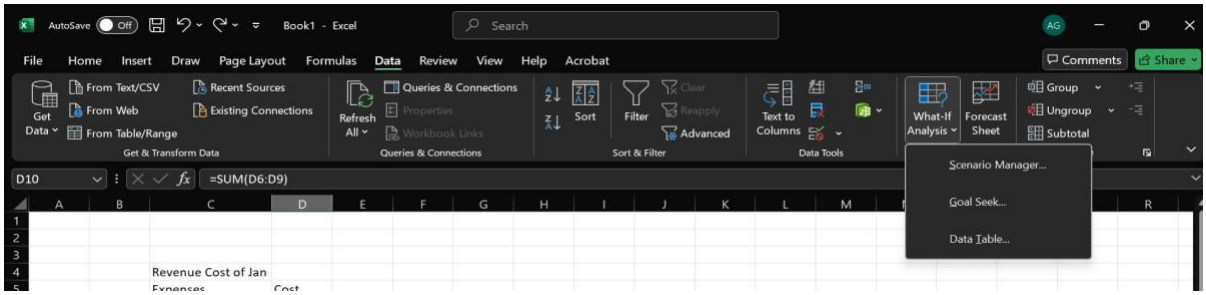
Step 1: Write all subjects and their marks in an excel sheet and do the **sum** by applying the formula sum.



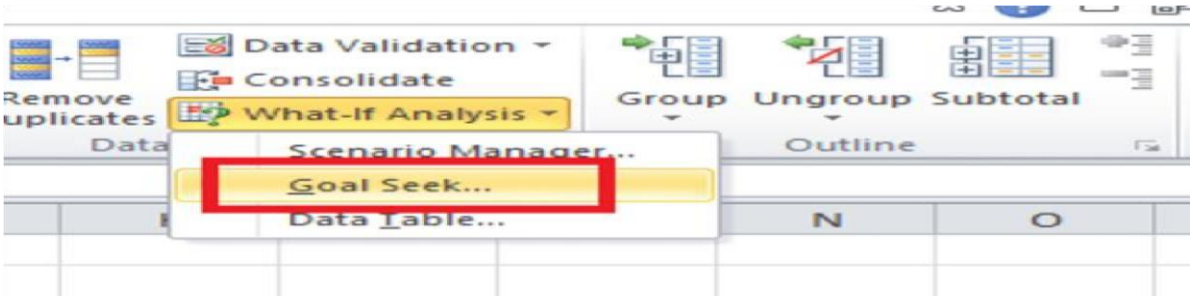
Step 2: Go into the **data** tab of the **Toolbar**.



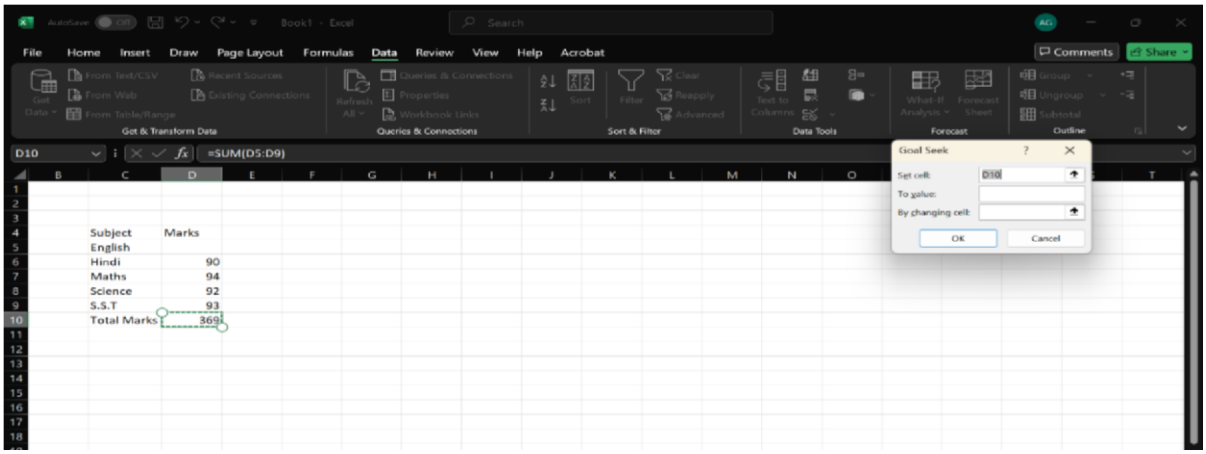
Step 3: Under the **Data Table** section, select the **What-if analysis**.



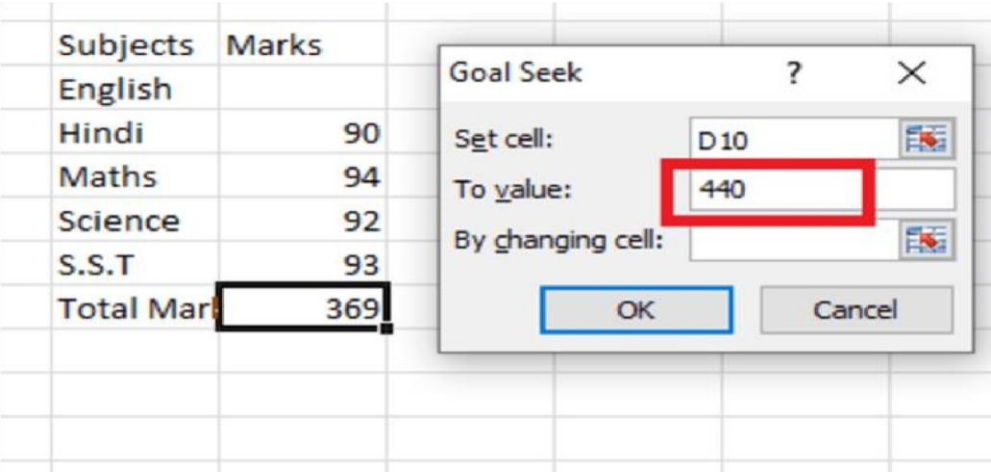
Step 4: A drop-down appears. Select the **Goal Seek.**



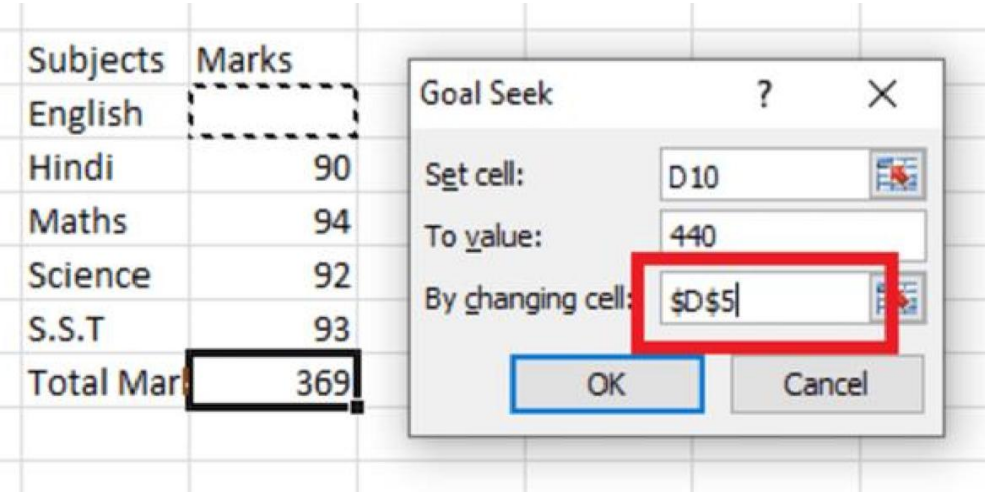
Step 5: The **dialogue box appears** in the first column write the name of the cell in which you apply the formula sum. Type **D10** in **Set cell**



Step 6: In the second column write the value of the **target.** The target value for this example is **440**.



Step 7: In the third column write the name of the cell in which you want to get marks in **English**. Provide absolute cell reference, i.e. **$D$5**.



Step 8: Click **ok** and see the result. The estimated marks for **English** are **71**.

Subject	Marks
English	71
Hindi	90
Maths	94
Science	92
S.S.T	93
Total Marks	440

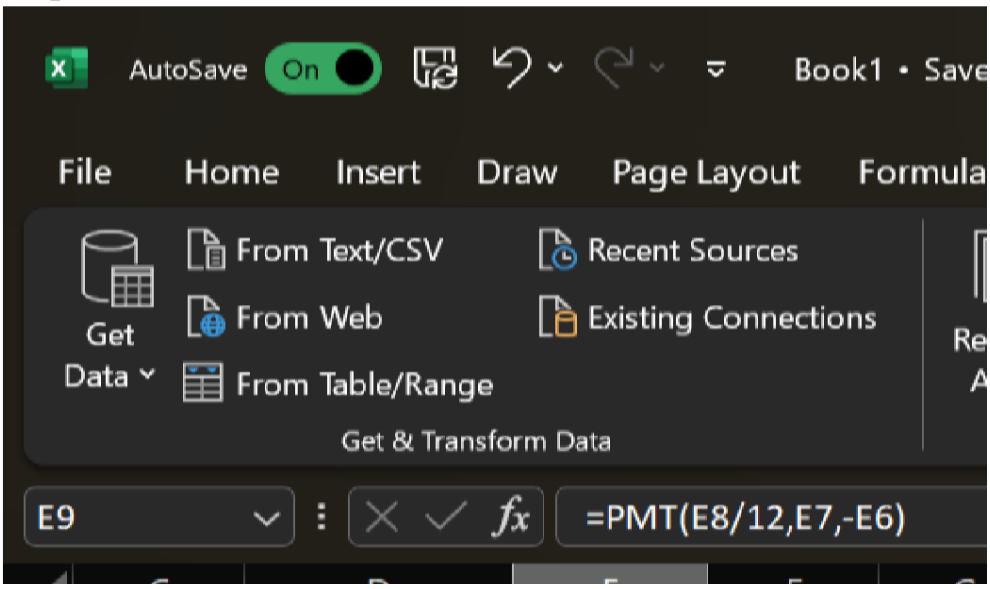
Data Table

In data, we create a table with different input values for the same variables. It is one of the most helpful features in what-if analysis. One can change different values in x and can achieve different outputs accordingly for research as well as business-driven purposes.

Data table in two Variable

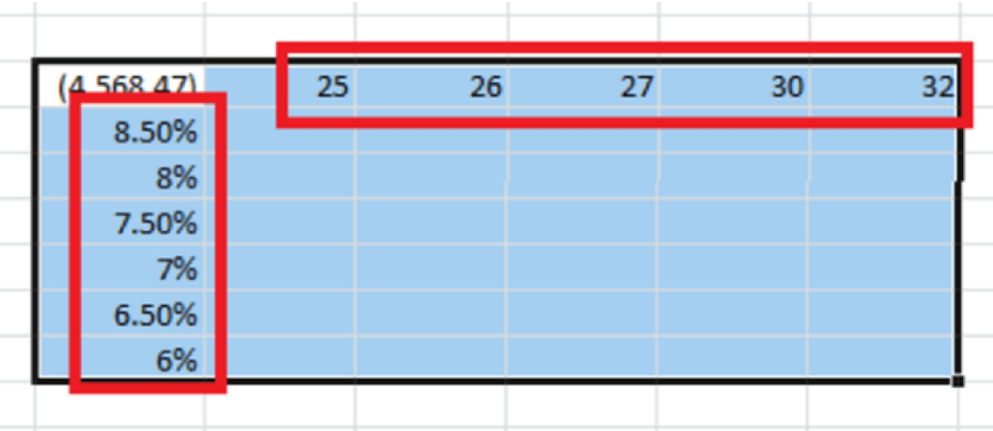
In the Data table in two variables, we can change **two input values in both row and column**. It includes two input cells. For example, A person wants to know about **per month instalments of loan** by the **different rates of interest and for the different time periods for the same principal amount.**

Step 1: Create a **table to find PMT.**

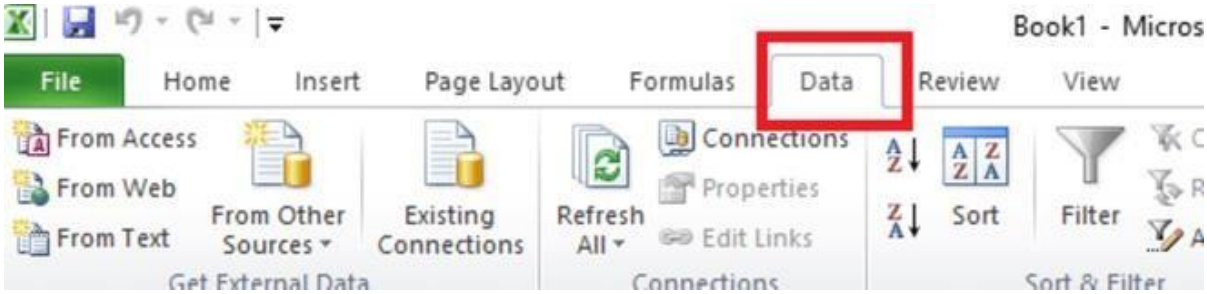


Step 2: Copy the last cell in which you get output in another cell.

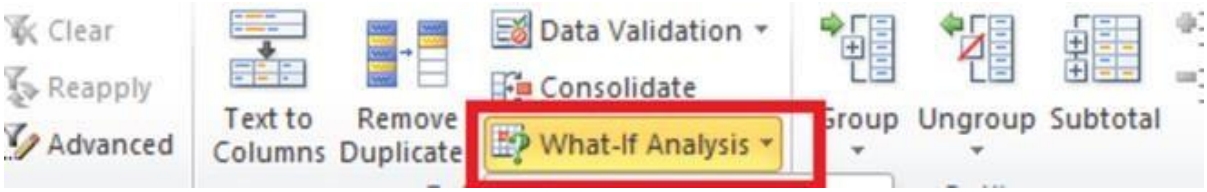
Step 3: Write both values you want to change in both **columns and rows.**



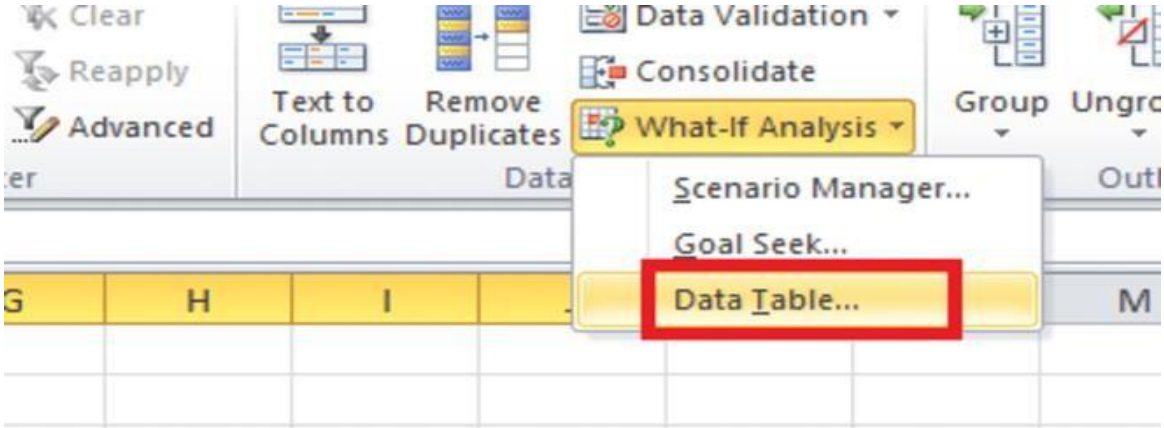
Step 4: Go to the **Data** tab of the **toolbar.**



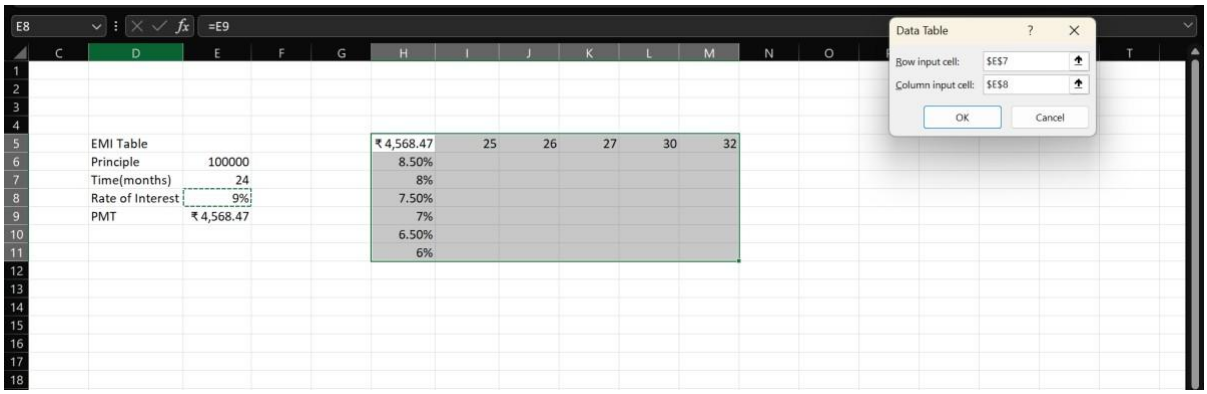
Step 5: Select the **what-if analysis.**



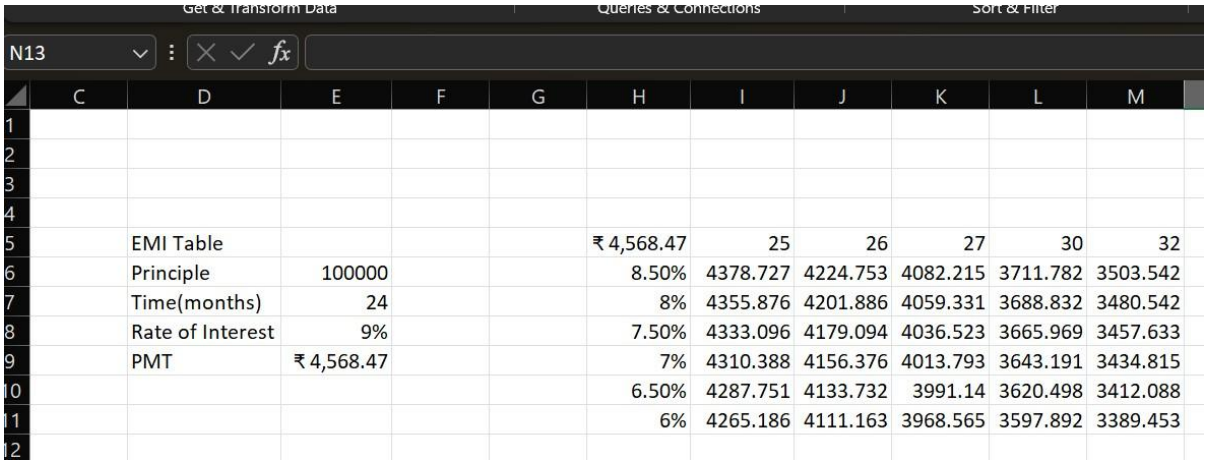
Step 6: Select the **Data Table.**



Step 7: A dialogue box appears in which you have to select the cell in which you want to change the value in both row and column. The **Row input cell value** is **$E$7** and the **column input cell value** is **$E$8**.



Step 8: Click **ok** and see the result.



EMI Table			₹4,568.47	25	26	27	30	32
Principle	100000		8.50%	4378.727	4224.753	4082.215	3711.782	3503.542
Time(months)	24		8%	4355.876	4201.886	4059.331	3688.832	3480.542
Rate of Interest	9%		7.50%	4333.096	4179.094	4036.523	3665.969	3457.633
PMT	₹4,568.47		7%	4310.388	4156.376	4013.793	3643.191	3434.815
			6.50%	4287.751	4133.732	3991.14	3620.498	3412.088
			6%	4265.186	4111.163	3968.565	3597.892	3389.453